Final Project Report

Date	30September 2026
TeamID	6ab4756eb5170a9b58b3f57e
ProjectName	**Script-Controlled ACL – Restrict Record Access Based on Field Value**
Maximum Marks	3 Marks

FINAL PROJECT REPORT

Script-Controlled ACL – Restrict Record Access Based on Field Value

ServiceNow Developer Instance

8-Phase Project Submission Report

Project Platform	ServiceNow Developer Instance
Application/Table	Institution Details
Security Mechanism	Access Control Lists (ACLs)
Team	1 Project Leader + 4 Team Members
Submission Structure	8 phases + GitHub repository + Google Drive demo video

Submission Alignment

This report is organized to match the required eight-phase project submission structure: Brainstorming & Ideation, Requirement Analysis, Project Design, Project Planning, Project Development, Project Testing, Project Documentation, and Project Demonstration. The GitHub repository is expected to be public and contain the project files phase-wise. The demonstration video is expected to be uploaded to Google Drive with access set to Anyone with the link can view.

Phase	Required Phase	Report Coverage
Phase 1	Brainstorming & Ideation	Problem identification, ideation and proposed field-based ACL solution.
Phase 2	Requirement Analysis	Functional, security, platform and data requirements.
Phase 3	Project Design	Architecture, Use Case, DFD, ER model and ACL design.
Phase 4	Project Planning	Team responsibilities and seven-day schedule.
Phase 5	Project Development	ServiceNow configuration, table creation, roles, records and ACL implementation.
Phase 6	Project Testing	Test cases, validation and screenshot evidence.
Phase 7	Project Documentation	Final report, user manual, ACL matrix, script and supporting files.
Phase 8	Project Demonstration	Complete screen-sharing video with explanation, working process and final output.

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PHASE 1 – Brainstorming & Ideation

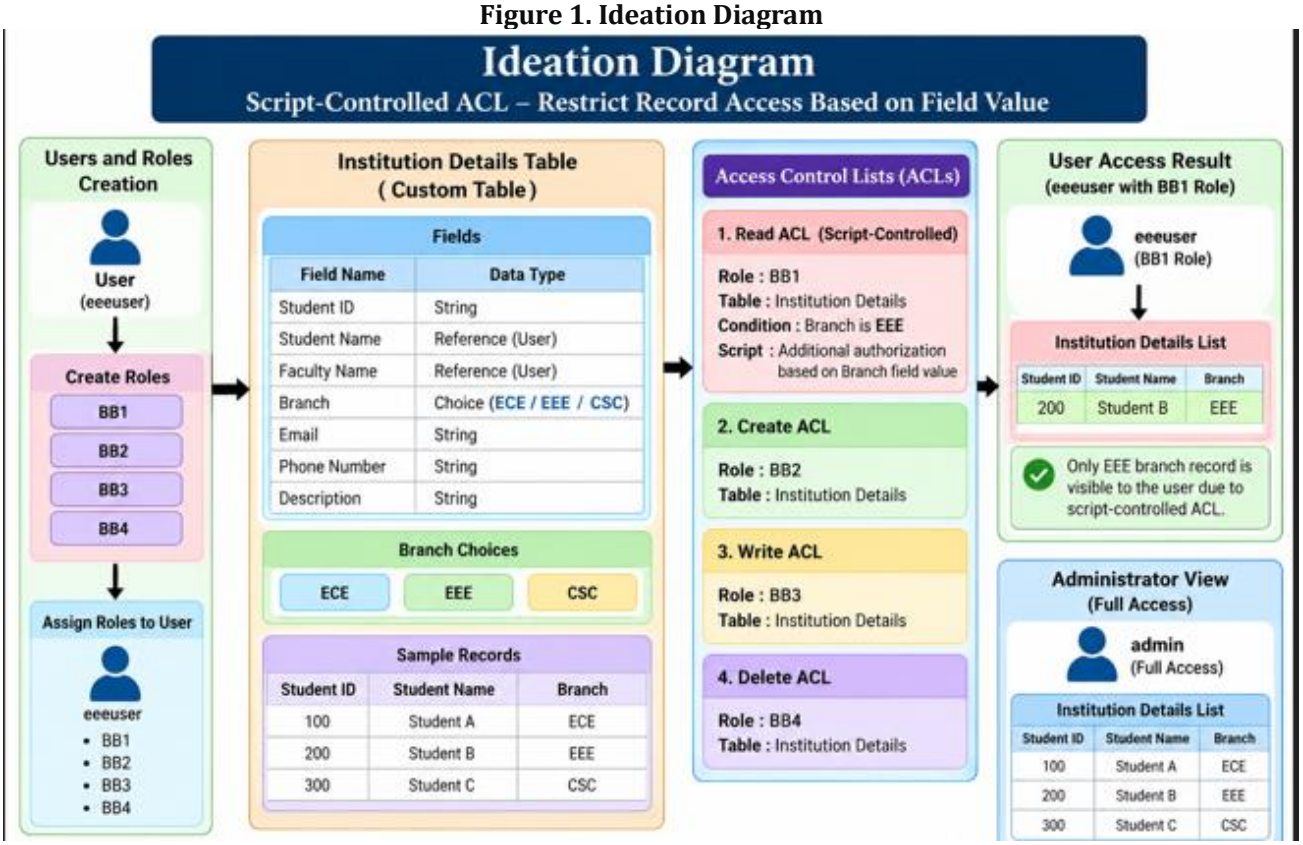
1.1 Project Overview

The project addresses the need to restrict ServiceNow record access according to a field value. The Institution Details table contains records for different branches, and the configured project user must not receive unrestricted access to records outside the authorized EEE branch.

1.2 Problem Statement

Role-based access alone may provide access to an entire table even when a user should only see records belonging to a particular branch. The proposed solution combines roles, ACL conditions and a script-based Branch check to provide record-level security.

Figure 1. Ideation Diagram



The ideation identifies the main security problem, the Institution Details data source, ACL processing, and the expected authorized/restricted result.

1.3 Objectives

- Create a project user and roles BB1–BB4.
- Create the Institution Details table and required fields.
- Configure ECE, EEE and CSC branch choices.
- Implement Read, Create, Write and Delete ACLs.
- Restrict Read access using the Branch value and authorization script.
- Test and document the complete project.

PHASE 2 – Requirement Analysis

2.1 Functional Requirements

- User creation and role assignment.
- Institution Details table creation.
- Branch choice configuration for ECE, EEE and CSC.
- Sample record creation.
- ACL configuration for Read, Create, Write and Delete.

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• Validation of authorized and restricted access.

2.2 Data Requirements

Field	Requirement
Student ID	String
Student Name	Reference
Faculty Name	Reference
Branch	Choice: ECE, EEE, CSC
Email	String
Phone Number	String
Description	String

2.3 Security Requirements

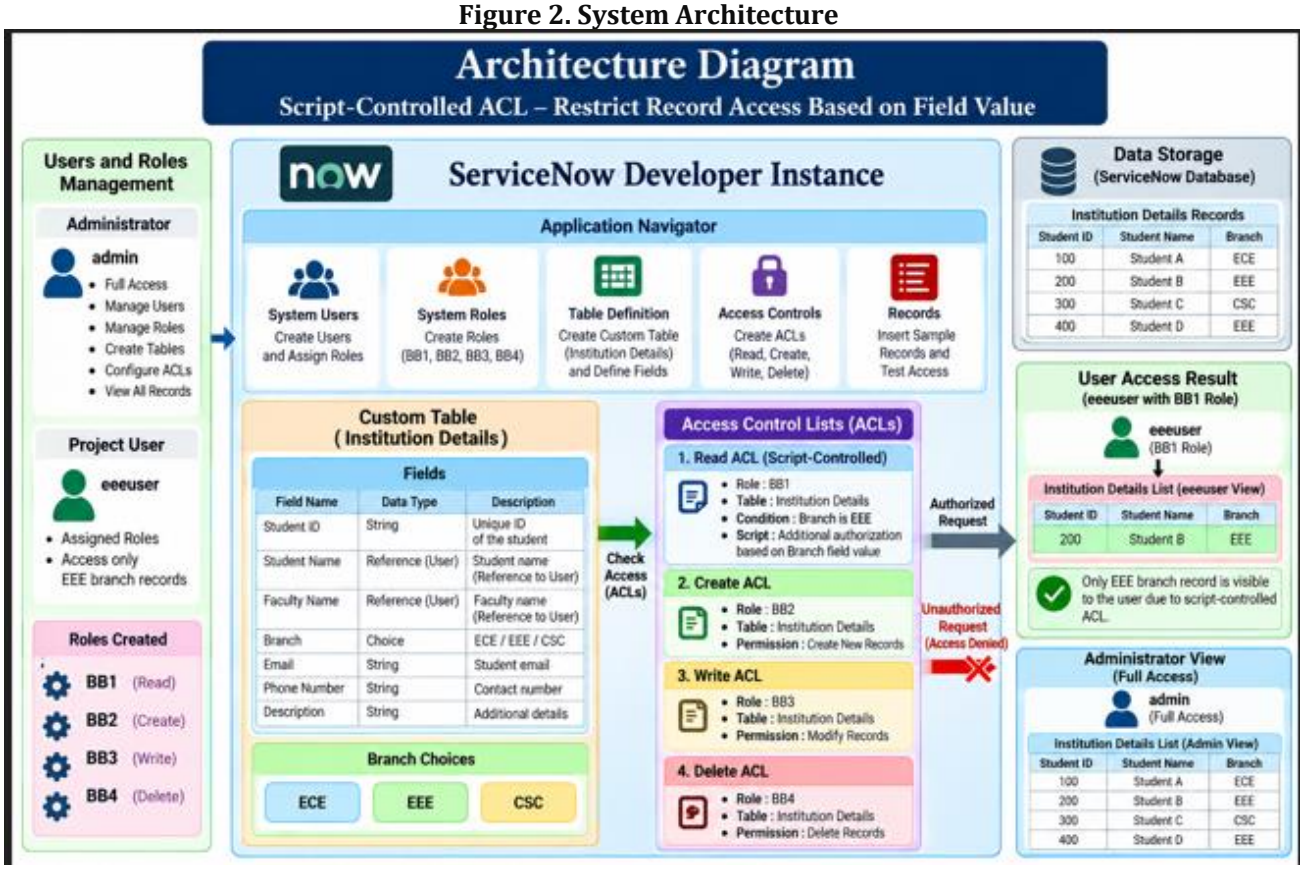
BB1 controls Read, BB2 controls Create, BB3 controls Write, and BB4 controls Delete. The Read ACL additionally applies the EEE Branch condition and script-controlled authorization.

PHASE 3 – Project Design

3.1 Architecture

The design connects users and roles to ACL processing, the Institution Details table and the resulting access decision.

Figure 2. System Architecture



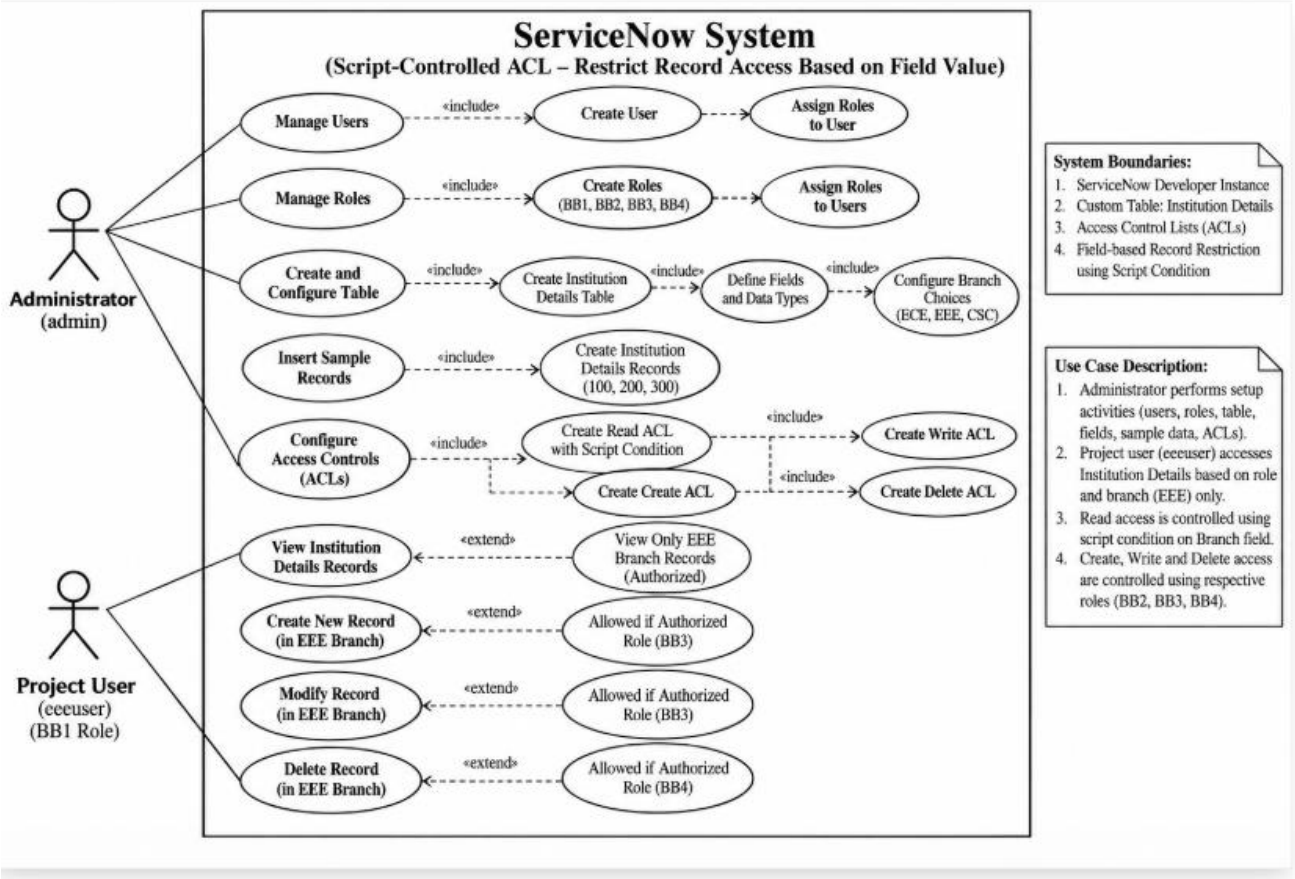
The architecture shows the user/role layer, ACL layer, Institution Details data, branch-based authorization and authorized versus restricted results.

3.2 Use Case

The Administrator configures the application and security rules; the Project User accesses records according to the configured roles and ACL rules.

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Figure 3. Use Case Diagram

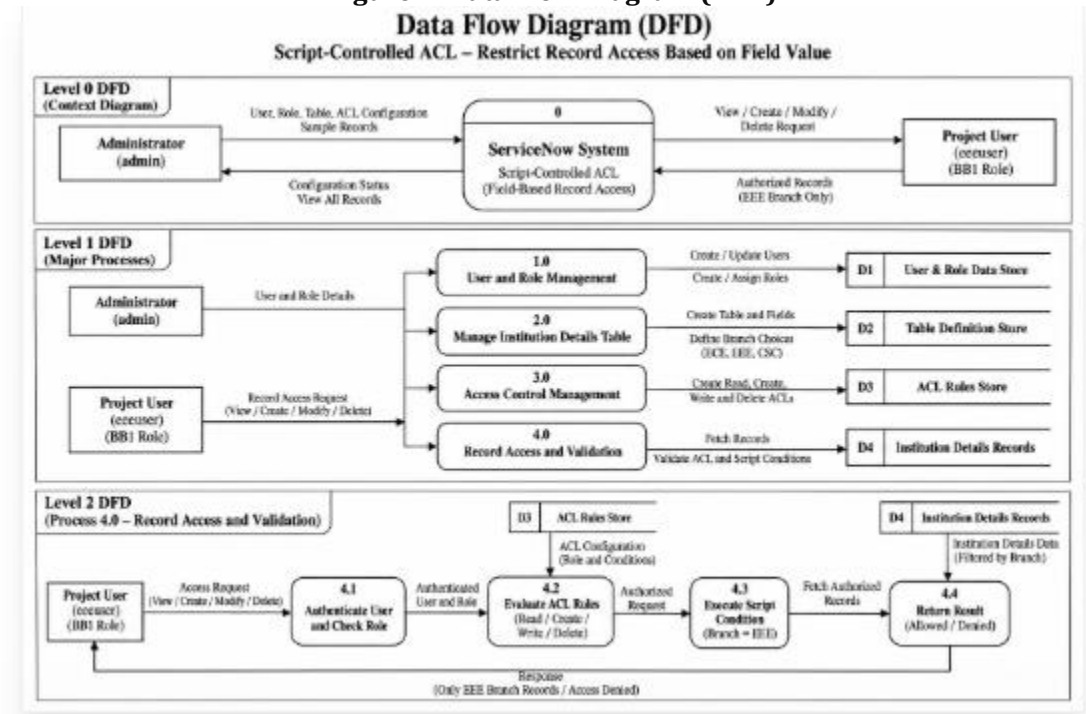


The use cases cover user/role management, table configuration, ACL configuration, record access and the final authorization decision.

3.3 Data Flow Diagram

The data flow begins with configuration and user access, passes through ACL evaluation, and ends with an authorized or restricted record response.

Figure 4. Data Flow Diagram (DFD)

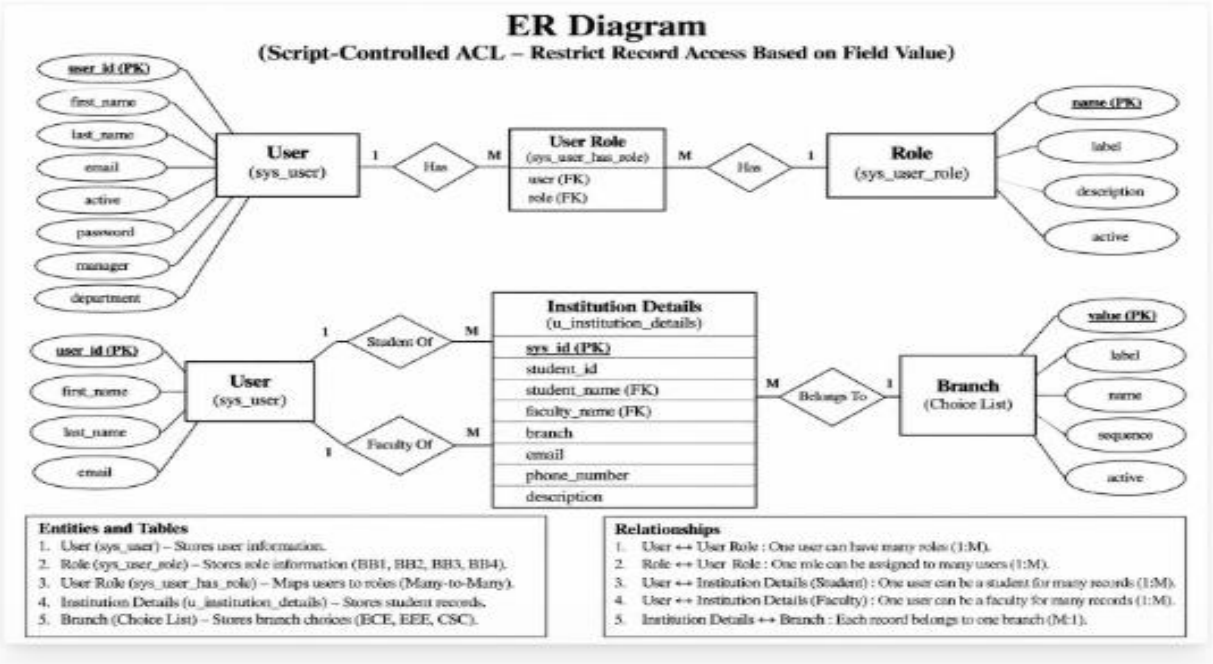


The DFD represents the flow from users and administrators through ServiceNow processes, ACL evaluation and record-level access results.

3.4 ER Diagram

The logical data model centers on Institution Details and relates user/role information to branch-based record security.

Figure 5. ER Diagram



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The ER model identifies the principal data entities and relationships used to support the Institution Details application and access-control design.

PHASE 4 – Project Planning

4.1 Team Roles

Member	Role	Main Responsibility
Al Aaffiya r	Project Leader	Main implementation, coordination, final submission and Google Drive video-link update.
Alex raj	Team Member	Documentation support.
Kathir	Team Member	Report preparation and formatting.
Jino	Team Member	Testing, validation and screenshots.
Aron roy	Team Member	Review, presentation and submission support.

4.2 Seven-Day Schedule

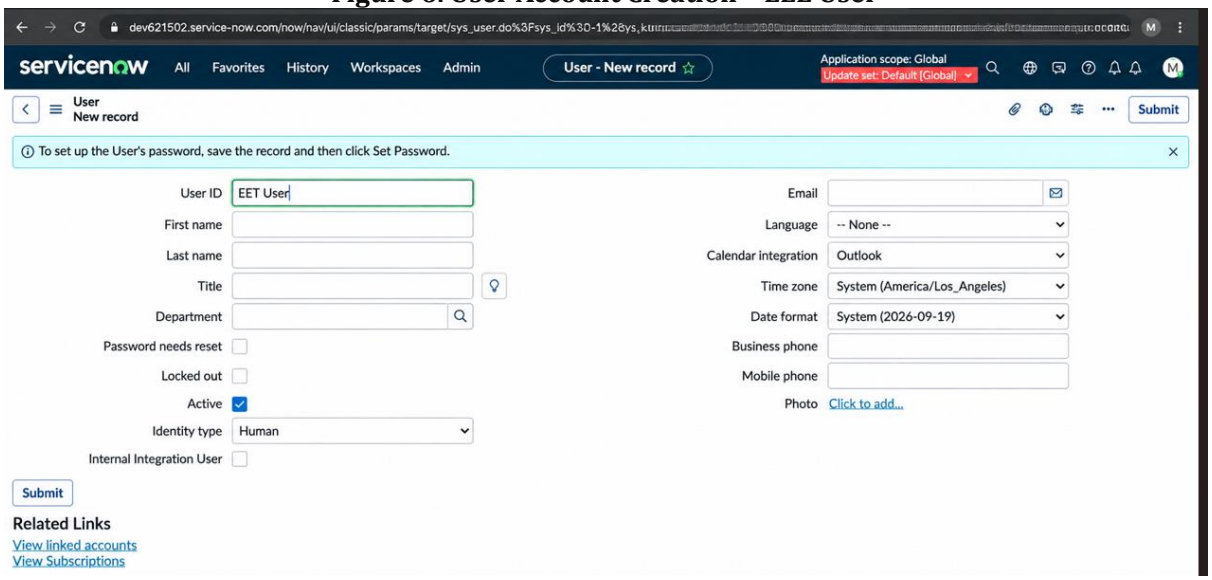
Day	Activity	Output
1	Planning	Requirements and responsibilities
2	Users, roles and table	Configuration started
3	Branches and records	ECE/EEE/CSC choices and sample records
4	ACL implementation	Read/Create/Write/Delete ACLs
5	Testing	Test results and screenshots
6	Documentation and video	Report and demonstration video
7	Final review/submission	Final package and Google Drive link

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PHASE 5 – Project Development

5.1 User Creation

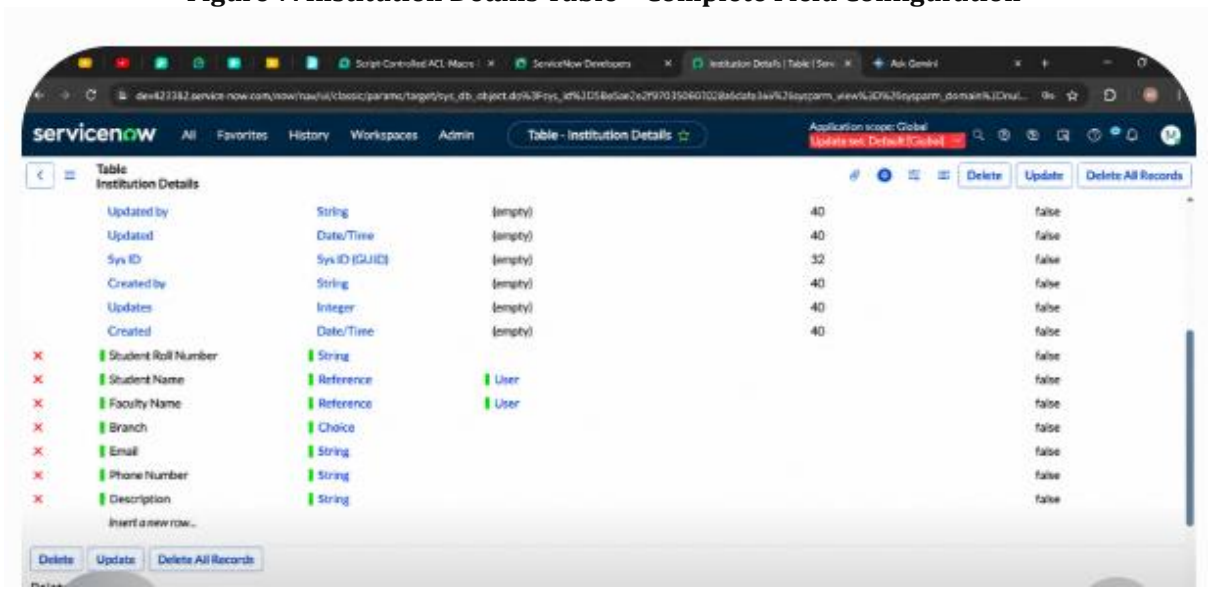
Figure 6. User Account Creation – EEE User



This screen demonstrates creation of the project test user used for ACL validation. The user is later evaluated against the configured role and branch-based access rules.

5.2 Table and Field Configuration

Figure 7. Institution Details Table – Complete Field Configuration

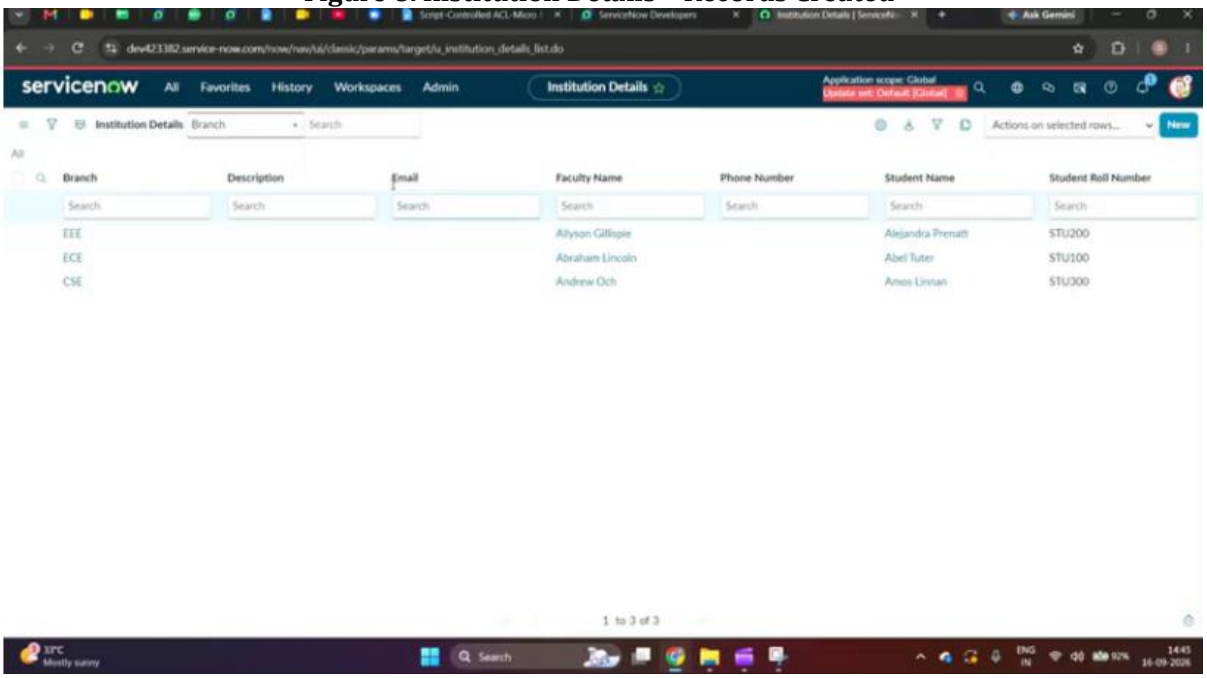


The table configuration provides the student, faculty, branch, email, phone and description fields required for the project. The Branch field is central to the Read access restriction.

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5.3 Sample Records

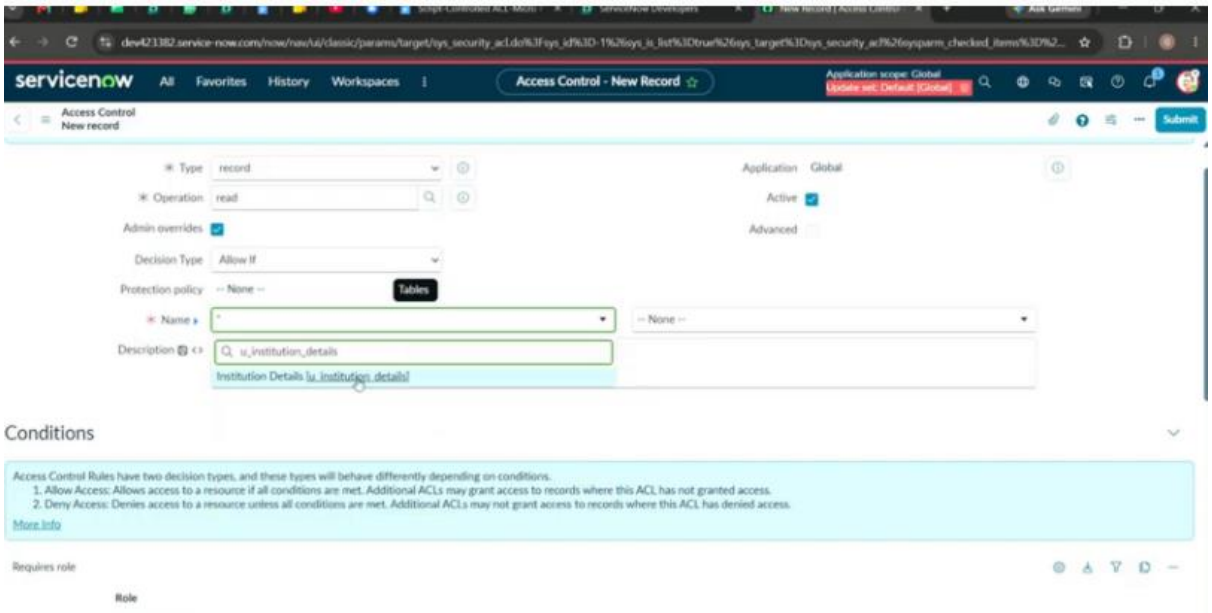
Figure 8. Institution Details – Records Created



Sample records are created to represent different branch values and provide controlled data for ACL testing. The records allow EEE authorization to be compared with ECE and CSC restrictions.

5.4 Read ACL

Figure 9. Read ACL – New Record for Read Operation

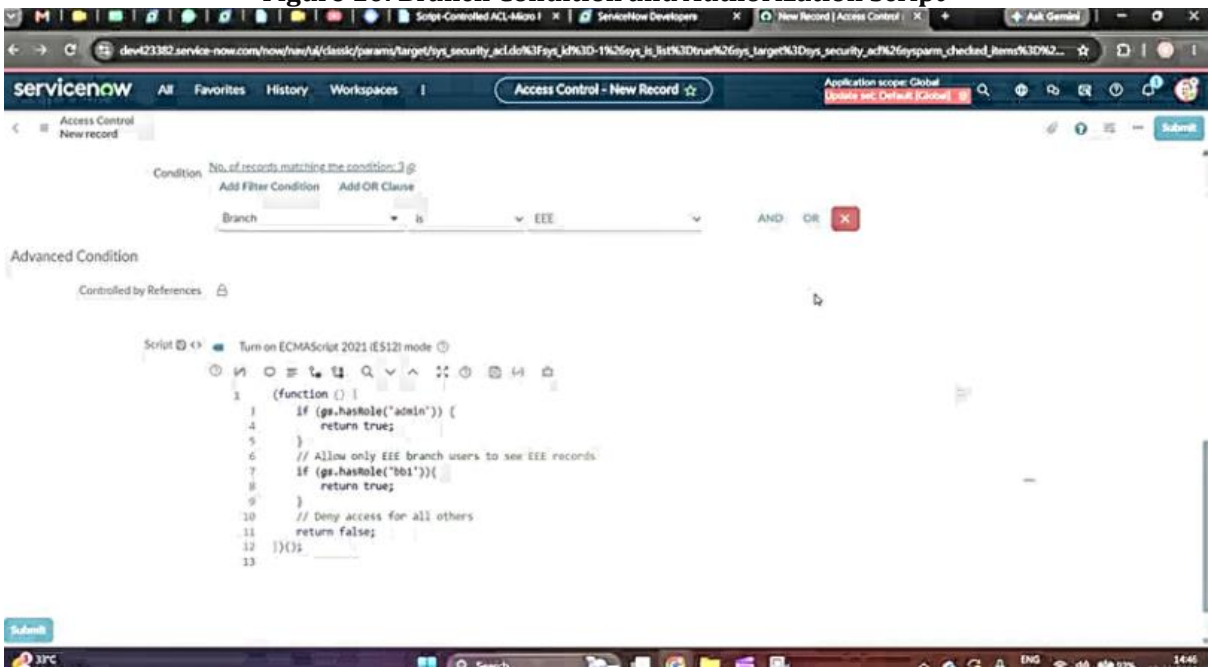


The Read ACL is created for the Institution Details table and is associated with the BB1 role. Its condition and script provide the field-based access restriction.

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5.5 Script-Controlled Authorization

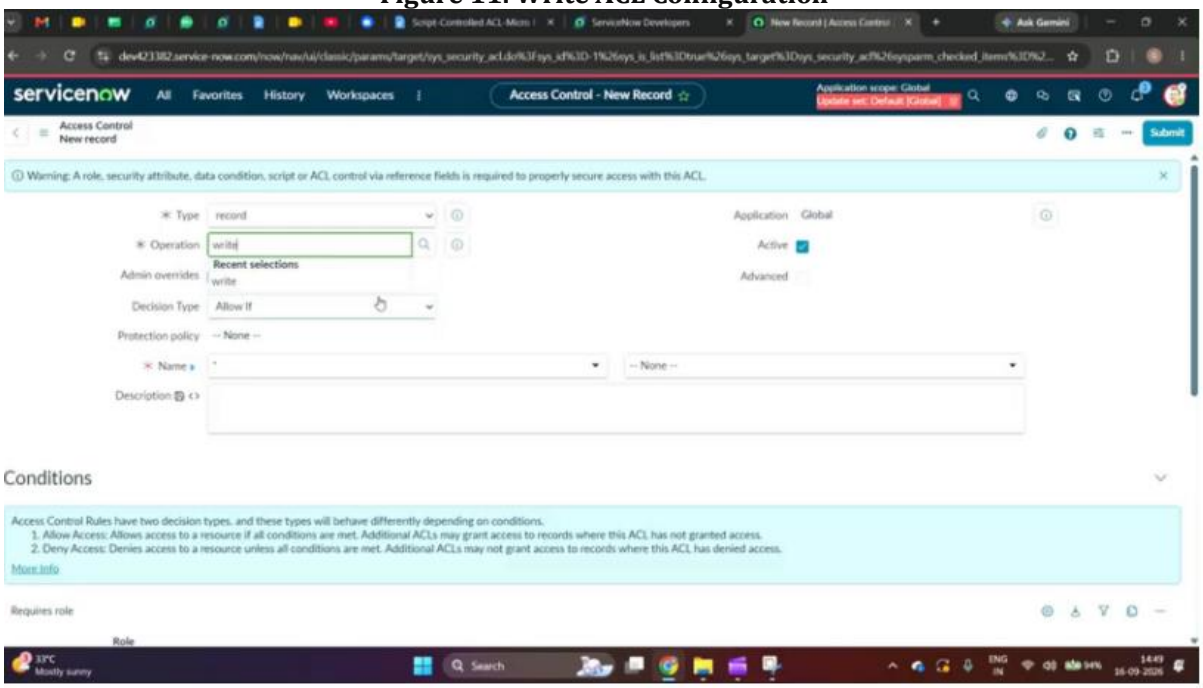
Figure 10. Branch Condition and Authorization Script



The ACL authorization section evaluates the current record's Branch value. The project rule permits the EEE branch and restricts other branch records for the configured Read user.

5.6 Create and Write ACLs

Figure 11. Write ACL Configuration

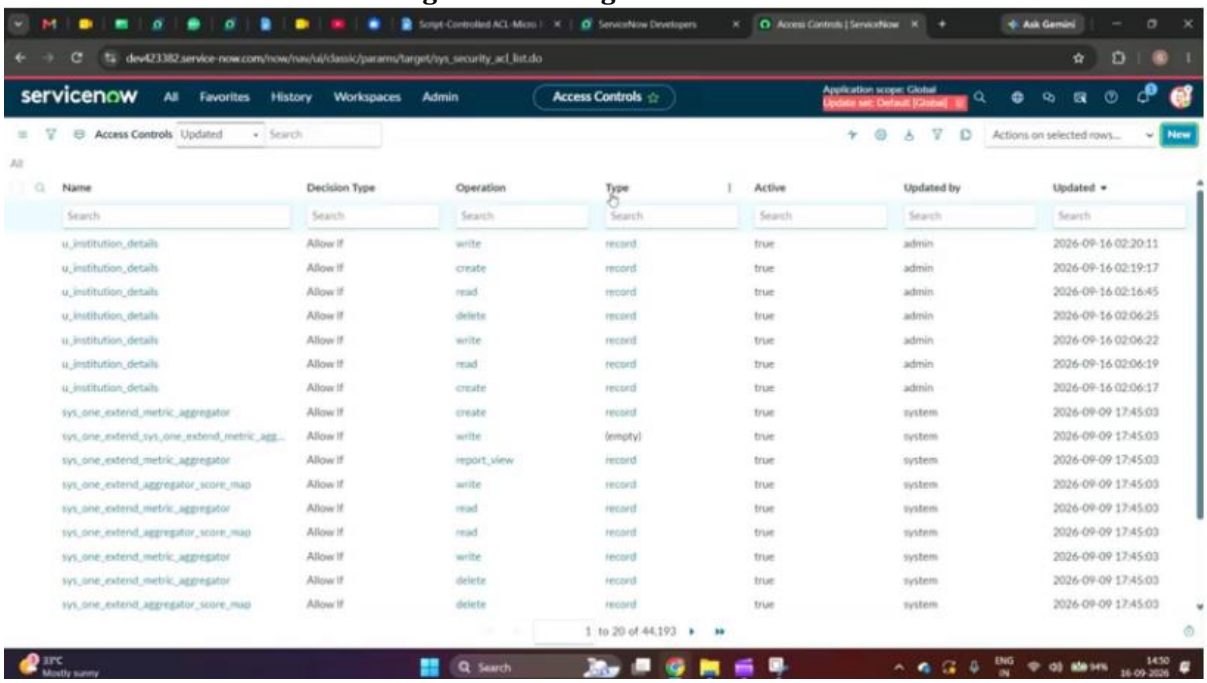


The Write ACL applies role-based authorization through BB3. It controls modification of existing Institution Details records.

5.7 ACL Verification

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Figure 12. Configured ACL Rules



This screen verifies the configured ACL rules for the Institution Details table. The project implements separate controls for Read, Create, Write and Delete operations.

PHASE 6 – Project Testing

6.1 Test Strategy

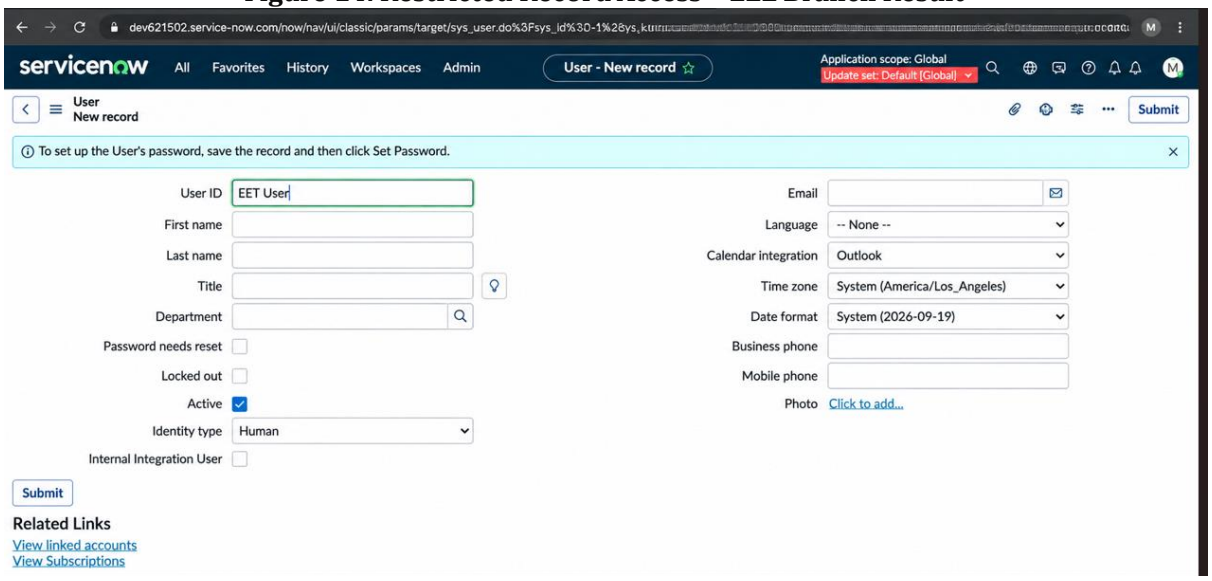
Testing verifies configuration, role permissions, field-based Read access and operation-specific ACL behavior.

Test ID	Test	Expected Result
TC-01	User creation	User created successfully
TC-02	Role creation	BB1–BB4 created
TC-03	Table configuration	Institution Details available
TC-04	Branch choices	ECE, EEE and CSC available
TC-05	Sample records	Test records available
TC-06	Read – EEE	EEE record accessible
TC-07	Read – ECE	ECE record restricted
TC-08	Read – CSC	CSC record restricted
TC-09	Create	BB2-controlled creation
TC-10	Write	BB3-controlled modification
TC-11	Delete	BB4-controlled deletion
TC-12	Final validation	Access behavior matches requirements

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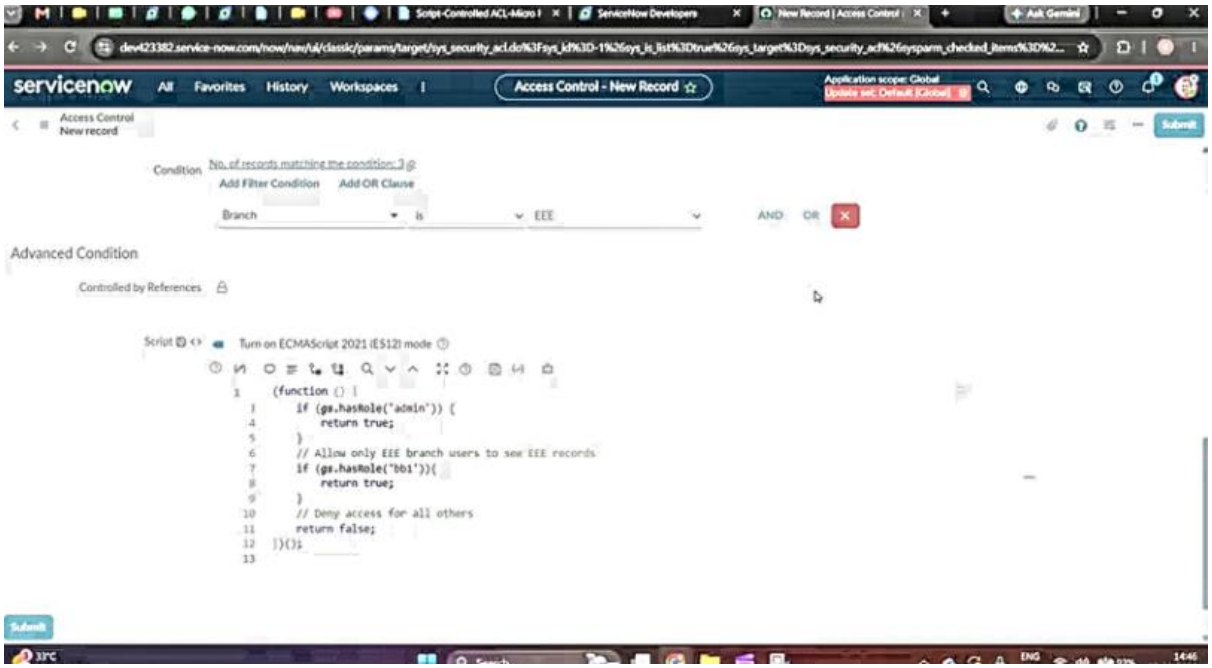
6.2 Final Access Validation

Figure 14. Restricted Record Access – EEE Branch Result



The validation screen demonstrates the intended field-based behavior: the configured user can see the permitted EEE branch record. This provides evidence that the Read ACL is being applied.

Figure 15. Administrator / Role-Based Access Validation



This evidence supports comparison between restricted project-user access and broader authorized administrative visibility. It is used to confirm the final security behavior.

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PHASE 7 – Project Documentation

7.1 ACL Matrix

ACL	Operation	Table	Role	Control
ACL-01	Read	Institution Details	BB1	Branch = EEE + script
ACL-02	Create	Institution Details	BB2	Role-based
ACL-03	Write	Institution Details	BB3	Role-based
ACL-04	Delete	Institution Details	BB4	Role-based

7.2 ACL Script

```
answer = (current.branch == 'EEE');
```

The script is placed in the Read ACL authorization section. It evaluates the current record's Branch value and returns true for the configured EEE value.

7.3 Documentation Package

- Final Project Report.
- User Manual.
- ACL Matrix and ACL Script.
- Update Set XML representation.
- Test Cases and Test Screenshots.
- Diagrams: Ideation, Architecture, Use Case, DFD and ER Diagram.
- Team Roles and 7-Day Project Schedule.

PHASE 8 – Project Demonstration

8.1 Demonstration Video Requirements

- Record a complete project demonstration using screen sharing.
- Include the student's voice-over/explanation.
- State the Project Name.
- Explain the Purpose and problem being addressed.
- Explain the Use/Benefits of the project.
- Demonstrate the project execution and working process.
- Show the final output and access-control result.
- Upload the completed video to Google Drive.
- Set the Drive sharing permission to Public/Anyone with the link can view.
- Submit the Google Drive demonstration link with the project.

8.2 GitHub Submission Requirements

The GitHub repository must be public and contain the project files phase-wise. A recommended structure is:

```
01_Brainstorming_and_Ideation/
02_Requirement_Analysis/
03_Project_Design/
04_Project_Planning/
05_Project_Development/
06_Project_Testing/
07_Project_Documentation/
08_Project_Demonstration/
README.md
```

8.3 Final Submission Checklist

- Public GitHub repository link is available.
- All eight phases are maintained in the repository.
- All project files/documents are uploaded.
- Final report contains the required figures and screenshots.
- Every figure has a descriptive title and two-line explanation.

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- Test screenshots support the test cases.
- Demo video includes project name, purpose, benefits, working process and final output.
- Google Drive video link is set to Anyone with the link can view.
- Project Leader completes the final submission and link update.

References

1. Project Document and project requirements supplied by the training/project team (provided in the user message).
2. Project Reference Video: “Script-Controlled ACL – Restrict Record Access Based on Field Value” supplied for the project.
3. ServiceNow Developer Instance configuration and screenshots captured from the supplied project workflow.

Note: The report intentionally uses the supplied project materials as the primary references. No external technical claims are required for the implementation description.

Appendix A – Project Evidence

Implementation screenshots and diagrams are embedded in the relevant phases above. The screenshot evidence follows the project workflow from user creation and table configuration through ACL creation, script authorization and final branch-based access validation.